Name :-
Roll # :- 24K-0573
Class :- BCS-4E

Date:

AI - Assignment 2

Q1 Part a

sol

- Domain : {P (Pit), G (Ghost), E (Exit)}
- Wind Strengths : Pit = Strong (S), Exit = Weak (W), Ghost = None
- Variables : $X_1, X_2, X_3, X_4, X_5, X_6$
- Measurement Rule : $Breeze(X_i, X_j) = max(Breeze(X_i), Breeze(X_j))$
  - if Pacman feels S, atleast one neighbour must be Pit
  - if Pacman feels W, must be Ghost and another be exit

a) Binary Constraint	Unary Constraint
$X_1 = P$ or $X_2 = P$, $X_2 = E$ or $X_3 = E$	$X_2 \neq P$
$X_3 = E$ or $X_4 = E$, $X_4 = P$ or $X_5 = P$	$X_3 \neq P$
$X_5 = P$ or $X_6 = P$, $X_1 = P$ or $X_6 = P$	$X_4 \neq P$
$\forall i,j$ s.t $Adj(i,j)$ $\neg(X_i = E$ and $X_j = E)$	

Part b

sol $X_1 = E$

- Breeze (1,6) = S and Breeze (1,2) = S then if $X_1 = E$ so $X_6 = P$
- Prune X_2 as no two adjacent variable can both be E
- Remove P from X_3 and X_4 as Breeze (2,3) and (3,4) are both W and as per constraint can't be P.
-

X_1			E
X_2		G	
X_3		G	E
X_4		G	E
X_5	P	G	E
X_6	P		

Date:

Part c

According to MRV we can pick eith X_1 or X_5 to assign first as they both have size = 1

Part d.

$X_6 = G, \quad X_1 = P \quad X_5 = P$

1) Constraint (6, 1) is $S \Rightarrow X_6 = G,\ X_1 = P$
2) Constraint (5, 6) is $S \rightarrow X_6 = G,\ X_5 = P$
3) Constraint (4, 5) is S $X_5 = P$ then $X_4 \in \{G, E\}$
4) Constraint (1, 2) is $S \Rightarrow X_1 = P$ then $X_2 \in \{G, E\}$
5) Constraint (2, 3) is W al least one of $\{X_2, X_3\}$ must be E
6) Constraint (3, 4) is W atleast one of $\{X_3, X_4\}$ must be E

So

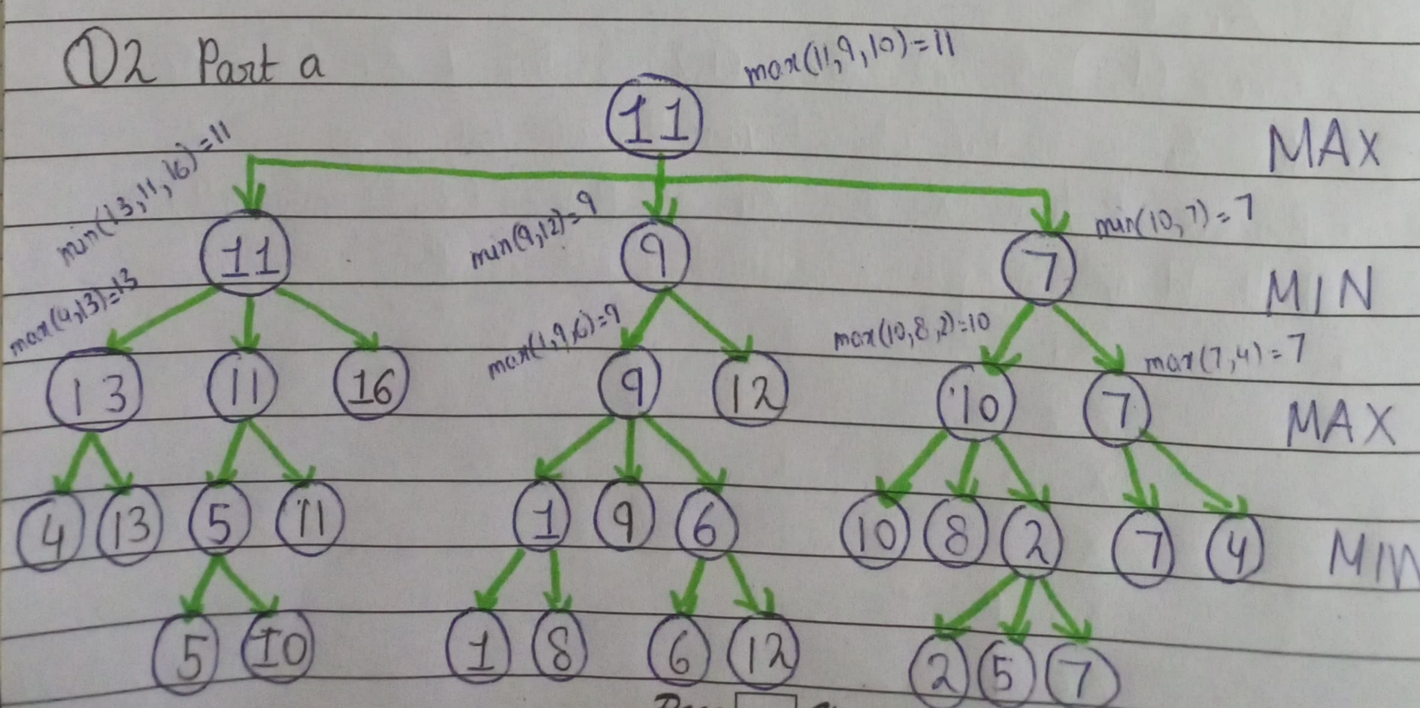
$X_3 = G$ both X_2 and X_4 are E

Sol: (P, E, G, E, P, G)

$X_3 = E$ so X_2 and X_4 are G

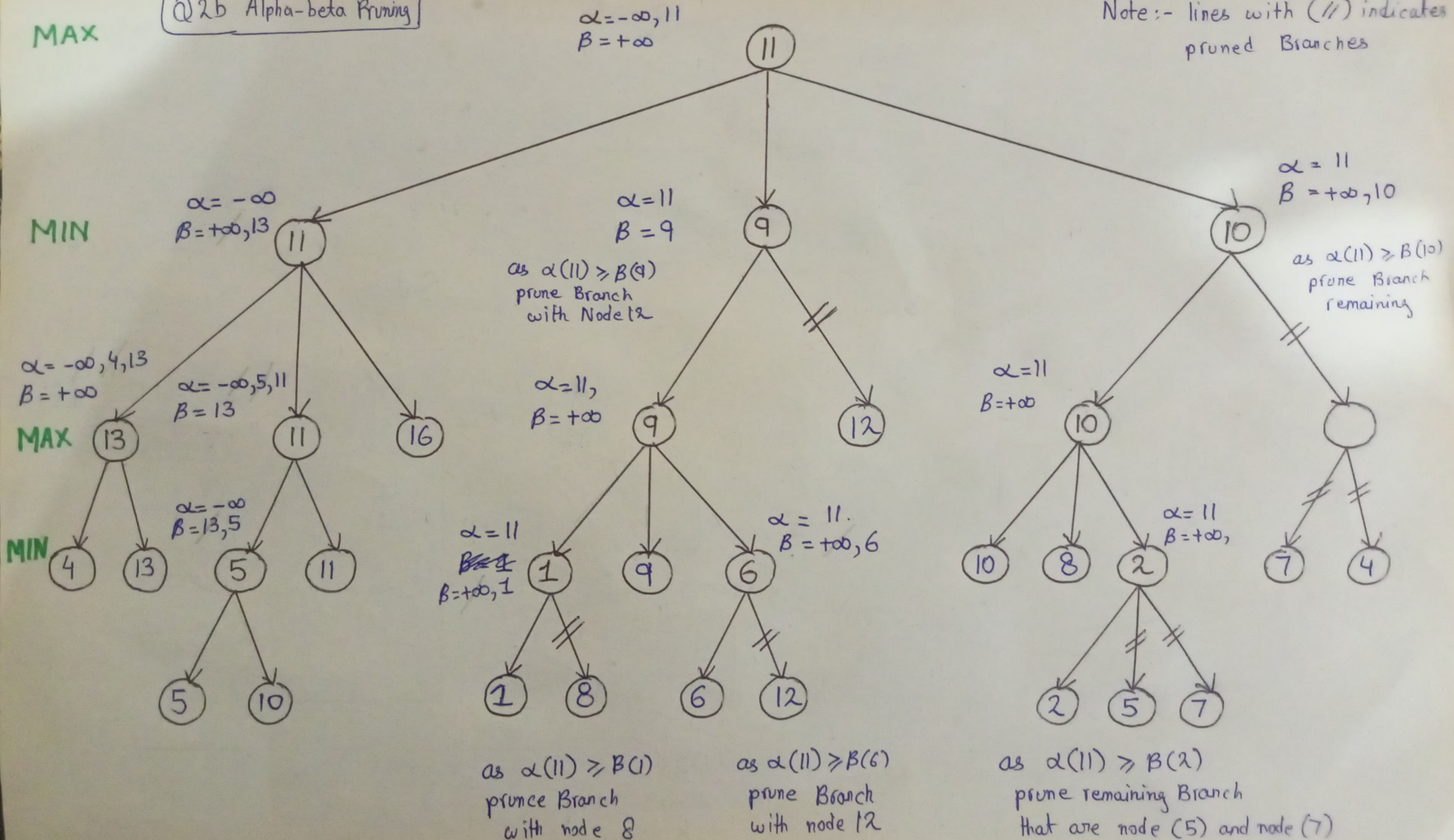
Sol (P, G, E, G, P, G)

Q2 Part a



Q2b Alpha-beta Pruning

Note :- lines with (//) indicates pruned Branches



Date:

Part ~~b~~c

No

Q3 Part a

Total Document 1 = 12 + 12 + 1 + 1 + 13 = 39 words

Total Document 2 = 17 + 17 + 0 + 0 + 17 = 51 words

Total Document 3 = 14 + 14 + 2 + 2 + 15 = 47 words

$$P(\text{Word}=\text{groot}) = P(\text{word}=\text{groot} \mid D=1)P(D=1) + P(\text{word}=\text{groot} \mid D=2)P(D=2) + P(\text{word}=\text{groot} \mid D=3)P(D=3)$$

- $P(D=1) = P(D=2) = P(D=3) = \frac{1}{3}$ (equal probability for all docs)

$$P(\text{word}=\text{groot}) = \left(\frac{13}{39} \times \frac{1}{3}\right) + \left(\frac{17}{51} \times \frac{1}{3}\right) + \left(\frac{15}{47} \times \frac{1}{3}\right)$$

$$= 0.3286$$

Part b

$$P(D=1 \mid \text{Word}=\text{we}) = \frac{P(W=\text{we} \mid D=1)}{P(\text{word}=\text{we} \mid D=1) + P(W=\text{we} \mid D=2) + P(W=\text{We} \mid D=3)}$$

$$= \frac{1/39}{1/39 + 0 + 2/47}$$

$$= 0.3760$$

Part c

$$P(\text{Word}=\text{am or } W=\text{are} \mid D=1) = 12+1/39 = 13/39 = 1/3$$

$$P(W=\text{am or } W=\text{are} \mid D=2) = 17+0/51 = 17/51 = 1/3$$

$$P(W=\text{am or } W=\text{are} \mid D=3) = 14+2/47 = 16/47$$

Date:

$$P(D=2 \mid W=\text{are or } W=\text{an}) = \frac{1/3}{1/3 + 1/3 + 16/47}$$

$$= 0.3310$$

Part d

$P(D=1) = 1/6 \qquad P(D=2) = 1/3 \qquad P(D=3) = 1/2$

$$P(\text{word} = \text{groot}) = \left(\frac{13}{39} \times \frac{1}{6}\right) + \left(\frac{17}{51} \times \frac{1}{3}\right) + \left(\frac{15}{47} \times \frac{1}{2}\right)$$

$$= 0.3262$$

Part e

$$P(D=1 \mid \text{word} = \text{we}) = \frac{\left(\frac{1}{6} \times \frac{1}{39}\right)}{\left(\frac{1}{6} \times \frac{1}{39}\right) + \left(\frac{1}{3} \times \frac{0}{51}\right) + \left(\frac{1}{2} \times \frac{2}{47}\right)}$$

$$= 0.1673$$

Part f

$$P(D=2 \mid \text{word} = \text{an or word} = \text{are}) = \frac{\frac{1}{3}\left(\frac{17}{51}\right)}{\left(\frac{1}{6} \times \frac{13}{39}\right) + \left(\frac{1}{3} \times \frac{17}{51}\right) + \left(\frac{1}{2} \times \frac{16}{47}\right)}$$

$$= 0.3298$$

Q4 Part a

$$P(R) = P(R \mid S) * P(S) + P(R \mid \neg S) * P(\neg S)$$

$$= 0.7(0.25) + 0.3(1 - 0.25)$$

$$= 0.7(0.25) + 0.3(0.75)$$

$$= 0.4 \qquad \text{ANSWER}$$

Date:

Part b

$P(We) = P(We|R,Wo) * P(Wo|R) * P(R) + P(We|R,\neg Wo) *$
$(1-P(Wo|R)) * P(R) + P(We|\neg R,Wo) * P(Wo|\neg R) *$
$(1-P(R)) + P(We|\neg R,\neg Wo) * (1-P(Wo|\neg R)) * (1-P(R))$
$= (0.12)(0.7)(0.4) + (0.25)(0.3)(0.4) + (0.1)(0.2)(0.6)$
$+ (0.08)(0.8)(0.6)$
$= 0.114$ ANSWER

Part c

$P(M|S) \Rightarrow$ The dependency chain $S \rightarrow R \rightarrow Wo \rightarrow We \rightarrow M$

Step 1 : Compute how likely the grass is wet given spring

$P(We|S) \Rightarrow$ cases

We depends on R and Wo

Wo depends on R

R depends on S

So we must consided all path from S to We

Combination

R	Wo
T	T
T	F
F	T
F	F

so

$P(We|S) = P(We|R,Wo) P(Wo|R) P(R|S) +$
$P(We|R,\neg Wo) (1-P(Wo|R)) P(R|S) +$
$P(We|\neg R,Wo) P(Wo|\neg R) (1-P(R|S)) +$
$P(We|\neg R,\neg Wo)(1-P(Wo|\neg R))(1-P(R|S))$
$= 0.12(0.7)(0.7) + (0.25)(0.3)(0.7) + (0.1)(0.2)(0.3)$
$+ 0.08(0.8)(0.3)$

Date:

$= 0.1365$

Now

$$P(M|S) = \frac{P(M,S)}{P(S)} = \frac{P(M|We)P(We|S)P(S) + P(M|\neg We)(1-P(We|S))P(S)}{P(S)}$$

$$= P(M|We)P(We|S) + P(M|\neg We)(1-P(We|S))$$

$$= 0.02(0.1365) + 0.42(1-0.1365)$$

$$= 0.3654 \quad \text{ANSWER}$$

Part d

We first need to find $P(Wo|S)$ as worm affects bird as well so

$$P(Wo|S) = P(Wo|R)P(R|S) + P(Wo|\neg R)P(\neg R|S)$$

$$= 0.7(0.7) + 0.2(1-0.7)$$

$$= 0.49 + 0.06$$

$$= 0.55$$

so

$P(Wo|S) = 0.55$ $\qquad$ $P(\neg Wo|S) = 0.45$

Now $P(B|S) = P(B|S,Wo)P(Wo|S) + P(B|S,\neg Wo)(1-P(Wo|S))$

$$= 0.8(0.55) + 0.4(0.45)$$

$$= 0.62 \quad \text{ANSWER}$$

Part e

Need to find $P(S|B,\neg Wo))$

$$P(S|B,\neg Wo) = \frac{P(B,S,\neg Wo)}{P(B,\neg Wo)}$$

Date:

Finding $P(Wo|\neg S)$ = if it's NOT spring whats the probab that worm appear

$$P(Wo|\neg S) = P(Wo|R)\,P(R|\neg S) + P(Wo|\neg R)\,(1 - P(R|\neg S))$$
$$= 0.7(0.3) + 0.2(1 - 0.7)$$
$$= 0.35$$

so

$P(Wo|\neg S) = 0.35$ $\quad P(Wo|S) = 0.45$ } from part d

$P(\neg Wo|\neg S) = 0.65$

if it ain't spring there is 65% chance there are no worms.

Now $P(B, \neg Wo, S) = P(S)\,P(\neg Wo|S)\,P(B|S, \neg Wo)$

- Probability that it's spring
- Probability that there are NO worms when given it's spring
- Given that it's spring and No worms whats the probab that birds are present.

so

$$P(B, \neg Wo, S) = 0.25(0.45)(0.4)$$
$$= 0.045$$

this means it is spring there are no worm and birds are present

Now $P(B, \neg Wo) = P(S)\,P(\neg Wo|S)\,P(B|S, \neg Wo) + P(\neg S)\,P(\neg Wo|\neg S)\,P(B|\neg S, \neg Wo)$

- It's spring no worms, bird present
- It's not spring, no worms, birds present
- Overall we get probability that there are birds no worm regardres of Spring or not

Date:

$$P(B|\neg W_0) = 0.25(0.45)(0.4) + (0.75)(0.65)(0.4)$$
$$= 0.24$$

Final

$$P(S|B,\neg W_0) = \frac{P(B,S,\neg W_0)}{P(B,\neg W_0)}$$
$$= \frac{0.045}{0.24}$$
$$= 0.1875 \quad \text{ANSWER}$$

Q5 Part a

It is worse for regular emails to be classified as spam than it is for spam email to be classified as regular email.
Hence applying p(spam) = 0.1 is sensible.

Total Regular email = 4	add-one smoothing
Total spam email = 8	$P(x=1 \mid class) = \frac{count(x=1)+1}{\text{total emain in class} + 2}$

cause binary feature

For Spam

- $P(study \mid spam) = \frac{0+1}{8+2} = \frac{1}{10}$

- $P(free \mid spam) = \frac{8+1}{8+2} = \frac{9}{10}$

- $P(money \mid spam) = \frac{4+1}{8+2} = \frac{5}{10}$

Date:

for Regular

- $P(\text{study} \mid \text{regular}) = \frac{3+1}{4+2} = \frac{4}{6} = \frac{2}{3}$

- $P(\text{free} \mid \text{regular}) = \frac{1+1}{4+2} = \frac{2}{6} = \frac{1}{3}$

- $P(\text{money} \mid \text{regular}) = \frac{1+1}{4+2} = \frac{2}{6} = \frac{1}{3}$

Part b

Given sentence "money for psychology study"

Features:

study = 1
free - 0
money = 1

Compute likelyhood for spam

$P(\text{feature} \mid \text{spam}) = P(\text{study} = 1 \mid \text{spam})\, P(\text{free} = 0 \mid \text{spam}) \cdot P(\text{money} = 1 \mid \text{spam})$

$= 0.1\ (0.1)\ (0.5)$

$= 0.005$

multiply by prior so $P(\text{spam}) \cdot P(\text{features} \mid \text{spam})$

$= 0.1\ (0.005)$

$= 0.0005$

Compute likelyhood for regular.

$P(\text{feature} \mid \text{regular}) = P(\text{study} = 1 \mid \text{regular})\, P(\text{free} = 0 \mid \text{regular})\, P(\text{money} = 1 \mid \text{regular})$

$= \frac{2}{3} \cdot \left(1 - \frac{1}{3}\right) \cdot \left(\frac{1}{3}\right)$

Date:

$P(\text{features} | \text{regular}) = 4/27$

multiply by prior

$P(\text{regular})\, P(\text{features} | \text{regular}) = (1 - P(\text{spam}))\, P(\text{features} | \text{regular})$

$= (1 - 0.1) \cdot (4/27)$

$= 0.9\,(4/27)$

$= 2/15$

Compute posterior probability

$$P(\text{spam} | \text{features}) = \frac{P(\text{spam})\, P(\text{features} | \text{spam})}{P(\text{spam}) P(\text{features} | \text{spam}) + P(\text{regular}) P(\text{features} | \text{regular})}$$

$$= \frac{0.0005}{0.0005 + 2/15}$$

$= 0.00373599$

≈ 0.003736 ANSWER

Part c

we want $p(\text{spam} | s) = 0.5$

as per formula

$$P(\text{spam} | s) = \frac{P(\text{spam}) P(s | \text{spam})}{P(\text{spam}) P(s | \text{spam}) + P(\text{regular}) P(s | \text{regular})}$$

we get

$p(\text{spam} | s) = 0.5$ when neumerator is equal to denominator

so from part b we have

$P(\text{feature} | \text{spam}) = (0.1)(0.1)(0.5) = 0.005$

$P(\text{feature} | \text{regular}) = (2/3)(1 - 1/3)(1/3) = 4/27$

Now let prior be p meaining $p(\text{spam}) = p$

Date:

To calc spam and regular score

$p(spam)\,p(feature|spam) = p(0.005)$

$= 0.005p$

$p(regular)\,p(feature|regular) = (1-p(spam))(4/27)$

$= \frac{4}{27} - \frac{4}{27}p$

Now as Spam Score = Regular Score

$0.005p = \frac{4}{27} - \frac{4}{27}p$

$\frac{827p}{5400} = \frac{4}{27}$

$p = \frac{800}{827}$

$p = 0.9673$ ANSWER

so $p(spam) = 0.9673$

① 6

Date:

Q6 let x = no. of crab
y = no. of clam

x	y	x^2	y^2	xy
2	137	4	18769	274
4	70	16	4900	280
2	184	4	33856	368
5	0	25	0	0
4	35	16	1225	140
0	297	0	88209	0
3	122	9	14884	366
5	1	25	1	5
1	253	1	64009	253
3	150	9	22500	450
$\Sigma x=29$	$\Sigma y=1249$	$\Sigma x^2=109$	$\Sigma y^2=248353$	$\Sigma xy=2136$

$n=10$
as we have 10 datas

slope formula

$$m = \frac{n(\Sigma(xy)) - (\Sigma x)(\Sigma y)}{n(\Sigma(x^2)) - (\Sigma x)^2}$$

$$m = \frac{10(2136) - (29)(1249)}{10(109) - (29)^2}$$

$$m = \frac{-14861}{249}$$

$m = -59.6827$

$m \approx -59.683$

Date:

calculate b (intercept) : $\frac{\Sigma y - m(\Sigma x)}{n}$

$= \frac{1249 - (-14861/249)(29)}{10}$

$= 297.9799$

≈ 297.98

so equation

$y = mx + b$

$y = -59.683x + 297.98$ ANSWER

Q7

Main Formula :- $P(Y=1) = \frac{1}{1+e^{-z}}$

where z:

$z = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_5x_5$ } sigmoid function

new data or new feature

Age (x_1) = 60

Diabetes (x_2) = 1 (yes)

Hypertension (x_3) = 1 (yes)

Smoker (x_4) = 1 (yes)

Surgery type (x_5) = 2 (major)

Assume the trained logistic regression mode produced these coefficients

Variable	
Intercept b_0	= -8
Age (b_1)	= 0.05
Diabetes (b_2)	= 1.5
Hypertension (b_3)	= 1.2
Smoker (b_4)	= 1.0
Surgery type (b_5)	= 2.0

Date:

Compute z

$$z = -8 + (0.05)(60) + (1.5)(1) + (1.2)(1) + (1.0)(1) + (2.0)(2)$$

$$z = 2.7$$

Probability $\Rightarrow$ $P(Y=1) = \dfrac{1}{1+e^{-2.7}}$

$= 0.937$ ANSWER

For a 60yr old patient with diabetes and hypertension, who smokes and is undergoing major surgery, the model predicts a very high probability (~93.7%) of post-surgery complication.